The Lovász number of random circulant graphs of even order

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Abstract

We extend the sharp expectation asymptotic of Chen, Hu, Hu, Wang and Yang from odd to even orders. In the random circulant graph on $\mathbb{Z}/n\mathbb{Z}$, every nonzero inverse orbit is selected independently with probability $1/2$, including the singleton orbit when n is even. We prove $\mathbb{E}\theta(G_n) = (1 + o(1))\sqrt{n}$ along the even integers; the known odd-order theorem then gives the limit through all integers. The extension retains the full Fourier operator but removes self-inverse residues from the canonical tree fields. An exact Walsh projection preserves the Wick-grafting estimate. The resulting leaf correction has squared Euclidean norm $4/n$. The Gaussian fixed-point and approximation theorems are imported from the odd-order paper, with all changes needed for the even-order transfer and expectation argument given here.

1 Statement and relation to previous work

For a finite simple graph G on n vertices, use the edge-zero convention

$$\theta(G) = \max\left\{\sum_{i,j} X_{ij} : X \succeq 0, \quad \operatorname{Tr} X = 1, \quad X_{ij} = 0 \text{ if } \{i,j\} \in E(G)\right\}.$$

Let G_n be the Cayley graph on $\mathbb{Z}/n\mathbb{Z}$ obtained by independently selecting each orbit $\{d, -d\}$, $d \neq 0$, with probability $1/2$. An orbit of size one receives one fair bit, just as an orbit of size two does.

Theorem 1.1. *For this model,*

$$\lim_{\substack{n\to\infty \\ n \text{ even}}} \frac{\mathbb{E}\theta(G_n)}{\sqrt{n}} = 1.$$

Consequently $\mathbb{E}\theta(G_n)/\sqrt{n} \to 1$ *through all positive integers.*

Bandeira et al. [1] obtained bounds within a $\log\log n$ factor of the square-root scale; the leading-constant question also appears in [2, Entry 9, Conjecture 18]. Chen et al. [3, Theorem 1.1] established the limit along all odd orders, including composite orders. We use that result for the odd subsequence. Our contribution is the even-order extension, with the independent orbit $\{n/2\}$ included. All numbered references to results of [3] refer to arXiv:2610.22971v1.

The proof uses three kinds of input from [3]: the independent-row Fourier sampling estimate from Section 5; the abstract comparison of Walsh moments in Theorem 7.2; and the Gaussian results in Theorem 9.3, Proposition 9.4 and Theorem 9.5. These inputs contain no restriction on an even cyclic modulus. We establish the modified tree estimates and the transfer back to the actual even-order graph below.

2 Fourier reduction and a deterministic certificate

Fix an even $n \geq 4$ and put $h = n/2$. Let $\epsilon_1, \ldots, \epsilon_h$ be independent fair signs, negative for selected edge orbits. Define

$$a_0 = 0, \quad a_d = a_{n-d} = \epsilon_d \ (1 \leq d < h), \quad a_h = \epsilon_h, \qquad F_{kd} = e^{-2\pi ikd/n}.$$

Write $D = \{d : a_d = -1\}$ and $N = \{d \neq 0 : a_d = 1\}$. On real vectors set

$$J = \frac{\mathbf{1}\mathbf{1}^{\mathsf{T}}}{n}, \qquad H_D = \frac{1}{n} F \operatorname{diag}(\mathbf{1}_D) F^*, \qquad H_N = \frac{1}{n} F \operatorname{diag}(\mathbf{1}_N) F^*, \qquad K = H_N - H_D.$$

These are real symmetric operators, and character orthogonality gives

$$J + H_D + H_N = I, \qquad K\mathbf{1} = 0, \qquad K^2 = I - J. \tag{1}$$

In particular $\|K\| \leq 1$. The multiplier a_h is retained in K.

A vector y is reflection-even if $y_k = y_{-k}$. Averaging a feasible matrix over cyclic shifts makes it circulant without changing its objective. Its Fourier eigenvalues give

$$\theta(G_D) = \max\{ny_0 : y \in \mathbb{R}^n \text{ reflection-even}, \ y \geq 0, \ \mathbf{1}^{\mathsf{T}} y = 1, \ (Fy)_D = 0\}. \tag{2}$$

For clarity, if the averaged matrix is $X_{ij} = x_{j-i}/n$, then $x_0 = 1$, $x_D = 0$, $Fx \geq 0$, and $y = Fx/n$, $x = Fy$ on the reflection-even subspace. This proves (2) for even n. Linear-program duality gives the complementary identity

$$\theta(G_D)\theta(G_N) = n. \tag{3}$$

The proof of [3, Propositions 3.2–3.3] uses only $F^*F = nI$, $F^2 = nI$ on reflection-even vectors, and the partition $\{0\} \sqcup D \sqcup N$; thus it applies here, including when one of D, N is empty. Complementation preserves our probability law, so

$$\mathbb{E}\theta(G_n) = \tfrac{1}{2}\mathbb{E}\left[\theta(G_n) + \frac{n}{\theta(G_n)}\right] \geq \sqrt{n}. \tag{4}$$

We record the deterministic certificate with its relevant derivation, so that its applicability to the singleton orbit is explicit. For $\tau > 0$ put

$$w_0 = 1, \quad w_k = \tau/n \ (k \neq 0), \quad W = 1 + \tau(1 - 1/n), \qquad \Phi_\tau(v)_k = \sqrt{v_k^2 + 4w_k}.$$

Lemma 2.1. *There are positive reflection-even vectors p^*, q^* with*

$$H_D p^* = H_N q^* = 0, \qquad p_k^* q_k^* = w_k, \qquad \sum p_k^* = \sum q_k^* = \sqrt{nW}.$$

Let $f^* = p^* - q^*$. For every real reflection-even mean-zero v, set $r = v - K\Phi_\tau(v)$ and

$$A_k = \frac{\Phi_\tau(v)_k + v_k}{2}, \qquad \Delta(v) = \sum_k w_k \left(\frac{A_k}{p_k^*} + \frac{p_k^*}{A_k} - 2 \right).$$

Then

$$0 \leq \Delta(v) \leq (\|v\|_2 + \tfrac{1}{2}\|f^*\|_2)\|r\|_2, \tag{5}$$

$$\left| \log \frac{\theta(G_D)}{\sqrt{n}} - \operatorname{arsinh}(v_0/2) \right| \leq \tfrac{1}{2} \log W + \sqrt{\Delta(v)}. \tag{6}$$

Proof. Minimize $(\sum p_k)^2/(2n) - \sum w_k \log p_k$ over positive reflection-even $p \in \ker H_D$. The domain contains $\mathbf{1}$; the Hessian $J + \operatorname{diag}(w_k/p_k^2)$ is positive definite. Sublevel sets keep the sum bounded above and away from zero, and then keep every coordinate away from zero. Hence the minimum exists and is unique. Stationarity with $q_k = w_k/p_k$ gives $q - Jp \in \operatorname{im} H_D$. The projection decomposition proves $H_N q = 0$, $Jq = Jp$ and $\langle p, q\rangle = (\sum p_k)^2/n = W$. This constructs the pair and also yields $f^* = K\Phi_\tau(f^*)$.

By (2), $p^*/\sqrt{nW}$ and $q^*/\sqrt{nW}$ are feasible for the complementary graphs. Using (3) and $p_0^* q_0^* = 1$ gives

$$p_0^*/\sqrt{W} \le \theta(G_D)/\sqrt{n} \le \sqrt{W}\, p_0^*.$$

If $u = \Phi_\tau(v)$, direct expansion using (1) gives

$$\Delta(v) = \sqrt{W/n}\sum_k u_k - 2W - \tfrac{1}{2}\langle r, f^*\rangle.$$

Put $s = (\sum u_k)/\sqrt{n}$. Since $\|u\|_2^2 = \|v\|_2^2 + 4W$,

$$s^2 - 4W = 2\langle v, r\rangle - \|r\|_2^2.$$

Dividing by $s + 2\sqrt{W}$ shows $\sqrt{W}(s - 2\sqrt{W}) \le \|v\|_2\|r\|_2$, proving (5). Finally $A_0 = e^{\operatorname{arsinh}(v_0/2)}$ and $e^z + e^{-z} - 2 \ge z^2$. The root term of $\Delta(v)$ therefore bounds $|\log p_0^* - \operatorname{arsinh}(v_0/2)|^2$. The preceding two-sided bound for theta proves (6). □

3 Tail integration and paired Fourier sampling

Put $b_0 = 1$ and $b_d = a_d$ for $d \ne 0$. Every feasible y in (2) satisfies $\langle y, Fb\rangle = ny_0$; hence $\theta(G_D) \le \max_k (Fb)_k$. Moreover

$$(Fb)_k = 1 + 2\sum_{d=1}^{h-1} \epsilon_d \cos(2\pi k d/n) + \epsilon_h(-1)^k.$$

The sum of squared coefficients of these independent signs is at most $4(h-1)+1 < 2n$. The bound $\cosh t \le e^{t^2/2}$ and a union bound give

$$\mathbb{P}\{\theta(G_n) > 1 + t\} \le n e^{-t^2/(4n)} \qquad (t \ge 0). \tag{7}$$

If an arbitrary event B has probability $p > 0$, integrate $\min(p, ne^{-t^2/(4n)})$ at $t = 2\sqrt{n\log(n/p)}$ to obtain

$$\mathbb{E}\left[\frac{\theta(G_n)}{\sqrt{n}}\mathbf{1}_B\right] \le \frac{p}{\sqrt{n}} + 2p\sqrt{\log(n/p)} + \frac{p}{\sqrt{\log(n/p)}}. \tag{8}$$

For $p = 0$ the expectation is zero. In particular it tends to zero whenever $p\sqrt{\log n} \to 0$.

Lemma 3.1. *There are absolute $C, c > 0$ such that, for all sufficiently large even n, with probability at least $1 - e^{-c(\log n)^2}$,*

$$\|x\|_2 \le \frac{C(\log n)^{3/2}}{\sqrt{n}}\|x\|_1 \quad \text{for every complex } x \text{ with } (Fx)_D = 0.$$

The same bound holds simultaneously for N after a change in c.

Proof. We give the modification to the transfer in [3, Lemma 5.3]. Take independent uniform rows $R_1, R_2, \dots$ of F, and an independent Poisson variable L of mean λn, where $\lambda = (\log 2)/2$. The occupation counts N_r are independent Poisson variables of mean λ. Condition on

$$H = \{N_0 = N_h = 0\}, \qquad \mathbb{P}(H) = e^{-2\lambda} = 1/2.$$

The other counts keep their original laws. Each ordinary reflected pair is occupied with probability $1 - e^{-2\lambda} = 1/2$, independently of the other pairs. Add h independently with probability $1/2$. This gives a set D' with exactly the law of D.

Let $q = \lceil n/4 \rceil$. Whenever $L \geq q$, all first q sampled rows lie in D' under H. If these rows satisfy restricted isometry on s-sparse vectors with parameter $\eta < 1$, the usual decomposition into successive blocks of the s largest coordinates yields

$$\|x\|_2 \leq \left(1 + \sqrt{\frac{1+\eta}{1-\eta}}\right) \frac{\|x\|_1}{\sqrt{s}} \quad (x \in \ker F_{D'}).$$

If the independent-row failure probability is δ, conditioning raises the failure bound to at most $2(\delta + e^{-c_* n})$, where $c_* = \lambda - 1/4 - \log(4\lambda)/4 > 0$ is the Poisson lower-tail constant. The independent-row estimate proved in [3, proof of Proposition 5.4] holds for every modulus: it uses bounded Fourier entries and column orthogonality, with no primality or parity hypothesis. It supplies fixed $\eta < 1$, $s \geq c_0 n/(\log n)^3$ and $\delta \leq e^{-c_1 (\log n)^2}$. This proves the bound for D. Complement symmetry and a union bound prove the simultaneous assertion. □

4 Canonical fields with the self-inverse residues excluded

Let T be an unordered rooted tree with at least two children at every internal vertex. Denote its node and leaf counts by $d(T)$ and $L(T)$. Assign $x_\ell \in \mathbb{Z}/n\mathbb{Z}$ to each leaf and define the flow $r_v(x) = \sum_{\ell \text{ below } v} x_\ell$. Call the assignment admissible when

$$r_v(x) \notin \{0, h\} \text{ for every } v, \qquad [r_v(x)] \text{ are pairwise distinct}, \tag{9}$$

where $[r] = \{r, -r\}$ is an unsigned label. Define

$$Z_T(k) = n^{-L(T)/2} \sum_{x \text{ admissible}} \cos\left(\frac{2\pi k r_{\text{root}}(x)}{n}\right) \prod_{v \in T} \epsilon_{[r_v(x)]}. \tag{10}$$

The notation $\epsilon_{[r]}$ means $\epsilon_{\min(r, n-r)}$. These real reflection-even fields have spatial mean zero and are independent of ϵ_h. Every $Z_T(k)$ is a homogeneous Walsh polynomial of degree $d(T)$ in the ordinary pair signs.

Proposition 4.1. *There is an absolute C_0 for which $A_D = \exp(C_0 D \log(D+2))$ has the following properties. For pairwise nonisomorphic trees of at most D nodes, coefficients c_T with $V = \sum_T \operatorname{aut}(T) c_T^2 \leq 1$, and $X(k) = \sum_T c_T Z_T(k)$,*

$$|\mathbb{E} X(k)^2 - V| \leq A_D n^{-1/2} \quad (k \notin \{0, h\}), \tag{11}$$

$$|\mathbb{E} e^{itX(k)} - e^{-Vt^2/2}| \leq A_D (t^2 + |t|^3) n^{-1/2} \quad (k \notin \{0, h\}), \tag{12}$$

$$\left| \mathbb{E} \frac{\|X\|_2^2}{n} - V \right| \leq A_D n^{-1/2}. \tag{13}$$

The corresponding two-mode comparison holds when both modes avoid $0, h$ and are unequal up to sign, with independent comparison Gaussian blocks. At every mode, without an exception,

$$\mathbb{E}|X(k)|^2 \le V D!\, 2^{3D+2}. \tag{14}$$

All constants are uniform when the degree and deterministic coefficients vary with n.

Proof. We describe the counting behind the extension of [3, Theorems 7.1 and 8.1, Proposition 8.2]. The row a_v of the leaf-incidence matrix A_T indicates the leaves below v. These sets are nested or disjoint. Subtracting selected descendant rows from each selected row leaves disjoint nonempty supports; choosing one leaf in each gives an identity minor. Thus prescribed values of selected rows S leave at most $n^{L-\rho(S)}$ assignments, for every modulus, where $\rho(S)$ is their real rank. Further,

$$\rho(S) + \rho(S^c) \ge L + 1 \quad (\varnothing \ne S \ne V(T)). \tag{15}$$

Indeed equality with L would make the two row spans disjoint. Each minimally supported relation $a_v = \sum_{u \text{ child of } v} a_u$ would then stay on one side. Connectivity would put all rows on the same side, a contradiction.

A fixed label set has at most $2^d d!$ oriented realizations. A prescribed label occurs in at most $2dn^{L-1}$ flows. Consequently second moments, maximal derivative influences and total influences are bounded, respectively, by P_D^2, P_D^3/n and P_D^3, for some $P_D = \exp(CD\log(D+2))$, uniformly in the output mode. For the derivative covariance $G_{ij} = d(T_i)^{-1}\sum_a (D_a Z_{T_i})(D_a Z_{T_j})$, remove its constant term and fix an overlap pattern. One tree, with leaves L_A, has a nonempty proper overlap set S. The number of full pairs is at most a degree-dependent constant times $n^{L_A+L_B-\rho(S)}$; fixing their outside Walsh support leaves at most that constant times $n^{L_A-\rho(S^c)}$ shared-label choices. Multiplying the maximum absolute coefficient by the sum of absolute coefficients, with the normalization $n^{-(L_A+L_B)/2}$ for each pair, gives an $O_D(n^{-1})$ sum of squared coefficients by (15). The number of role, orientation and matching patterns is $\exp(O(D\log(D+2)))$. Hence

$$\operatorname{Var}(G_{ij}) \le \frac{\exp(CD\log(D+2))}{n}.$$

Deleting assignments that use h only decreases these absolute counting bounds.

For the target covariance, matching admissible label sets determine a unique signed node bijection, since neither permitted residue is self-inverse. A bijection not preserving the real flow space forces a nonzero congruence with coefficient of magnitude at most D. The bound $\#\{x : cx = b \pmod n\} \le |c|$ gives at most Dn^{L-1} assignments. A preserving bijection supplies all n^L assignments before exclusions. The exclusions $r_v = 0$, $r_v = h$ and $r_u = \pm r_v$ remove $O(D^2)n^{L-1}$ assignments: each equation has a coefficient 1 or -1. If the leaf counts differ, the simpler bound $2^D D! n^{\min(L,L')}$ for matching pairs gives normalized error $O_D(n^{-1/2})$.

For a root-preserving bijection, unrestricted cosine orthogonality gives $1/2$ when the modes are equal up to sign and avoid $0, h$, and zero for distinct nonconjugate modes. The root-preserving signed bijections are a common sign times a rooted-tree isomorphism. To see this, the minimal row relations containing the root correspond to frontiers partitioning all leaves. Each proper node belongs to such a frontier, forcing every sign to agree with the root's sign. The frontier of all leaves has uniquely largest size, so the remaining permutation preserves leaves and their descendant-set hierarchy. There are therefore $2\operatorname{aut}(T)$ such bijections for T itself. When a root is matched with a proper subtree, one leaf inside and one outside give a determinant-one minor; the observed sums are independent uniform residues and their nonzero-mode cosine average vanishes. This proves the covariance target in (11). At h, the root-preserving average is 1, so that mode is exceptional.

The exchangeable sign-flip calculation of [3, Theorem 7.1] uses just the influence and derivative-covariance bounds above. Its sign count is now $h-1$; it cancels from the regression identity exactly as before. It gives (12) and its two-mode version. There are at most 4^D tree types, and $\sum|c_T| \le 2^D\sqrt{V}$, so enlarging C_0 absorbs all coefficient sums and the two exceptional coordinates into (13). Finally the absolute label-matching bound gives $\mathbb{E}|Z_T(k)|^2 \le 2^{d(T)}d(T)!$ at every k. The triangle and Cauchy–Schwarz inequalities, with at most 4^{D+1} trees, give (14). □

5 An exact projection and quantitative Wick grafting

Let He_r be the probabilists' Hermite polynomial, defined by $e^{tx-t^2/2} = \sum_{r\ge 0}\mathrm{He}_r(x)t^r/r!$. For distinct tree types T_i and integers $r_i \ge 0$ with $r = \sum r_i \ge 2$, graft a new root above r_i copies of each T_i to obtain T. Set

$$W(k) = \prod_i \frac{\mathrm{He}_{r_i}(Z_{T_i}(k)/\sqrt{\mathrm{aut}(T_i)})}{\sqrt{r_i!}}, \qquad d = 1 + \sum_i r_i d(T_i).$$

Let $\Pi_{d,\neg h}$ be orthogonal projection, in the full sign probability space, onto Walsh monomials of degree d which do not contain ϵ_h.

Lemma 5.1. *For every even n and every such collection of trees,*

$$\Pi_{d,\neg h}(KW) = \frac{Z_T}{\sqrt{\mathrm{aut}(T)}}. \tag{16}$$

Consequently

$$\mathbb{E}\frac{\|KW - Z_T/\sqrt{\mathrm{aut}(T)}\|_2^2}{n} = \mathbb{E}\frac{\|KW\|_2^2}{n} - \mathbb{E}\frac{\|Z_T\|_2^2}{n\,\mathrm{aut}(T)}. \tag{17}$$

Proof. Negating every leaf residue in (10) leaves its sign product unchanged, so the cosine can be replaced by $e^{-2\pi i k r_{\mathrm{root}}/n}$. Multiply the exponential forms for the distinguished child copies. Applying K imposes, by Fourier orthogonality, that the new root residue is the sum of the child-root residues. A zero root is killed by $a_0 = 0$. A root equal to h contributes ϵ_h, absent from every child, and is removed by $\Pi_{d,\neg h}$. For any other root, degree d requires all child-node and root labels to be distinct. These are precisely the admissible flows of T. A repeated label lowers the reduced Walsh degree by at least two. Every lower Hermite term already loses at least two positive-degree child factors, so after application of K it still has degree below d. The leading normalization is correct because

$$\mathrm{aut}(T) = \prod_i r_i!\,\mathrm{aut}(T_i)^{r_i}.$$

This proves (16); coordinatewise orthogonality and summation give (17). □

Proposition 5.2. *If $d(T_i) \le D$ and $2 \le r \le M$, then*

$$\mathbb{E}\frac{\|KW - Z_T/\sqrt{\mathrm{aut}(T)}\|_2^2}{n} \le \frac{\exp(C[(D+1)(M+1)]^4)}{\sqrt{n}}. \tag{18}$$

Let G_T be independent centered Gaussians with variance $\mathrm{aut}(T)$, and let J_{tr} send normalized Gaussian Wick monomials to the normalized coordinate of the grafted tree. For a linear combination $X = \sum c_T G_T$ of trees of at most D nodes, all with even internal arities, let $X[n] = \sum c_T Z_T$. If P is even of degree at most M and $Y[n]$ represents $J_{\mathrm{tr}}(P(X) - \mathbb{E}P(X))$, then

$$\mathbb{E}\frac{\|KP(X[n]) - Y[n]\|_2^2}{n} \le \mathbb{E}P(X)^2\frac{\exp(C[(D+1)(M+1)]^4)}{\sqrt{n}}. \tag{19}$$

Proof. The abstract Walsh moment comparison of [3, Theorem 7.2], applied with the influence and covariance bounds in the preceding proof, gives for any ordinary polynomial R of degree m in standardized tree fields at one generic mode or two generic nonconjugate modes

$$|\mathbb{E}R(Z) - \mathbb{E}R(G)| \le |R|_{\mathrm{coef}} \frac{e^{C(D+m+2)^4}}{\sqrt{n}}.$$

Here $|R|_{\mathrm{coef}}$ is its absolute coefficient sum; G consists of the corresponding independent standard normals. At arbitrary modes, sign hypercontractivity instead gives $\mathbb{E}|R(Z)| \le |R|_{\mathrm{coef}} e^{C(D+m+2)^4}$. For explicit growth control, the moment comparison uses an L^2 bound $B = P_D$, maximal influence $u = P_D^3/n$ and derivative covariance deviation $v = P_D^5/\sqrt{n}$, with $\log P_D = O(D\log(D+2))$. Its error terms $B^{m-2}v$ and $B^{m-1}\sqrt{u}$, times the degree factor $(32m)^{(D+6)m}$, are absorbed by the displayed exponential. Polarization transfers scalar moments to arbitrary monomials.

The Hermite coefficient formula gives $|W|_{\mathrm{coef}} \le e^{3r/2}\sqrt{r!}$. Apply the polynomial estimate to $W(k)^2$ and $W(k)W(l)$, of degree at most $2M$. Their Gaussian means are respectively one and zero for independent blocks. There are at most $6n$ exceptional ordered pairs: at most $4n$ containing 0 or h, and at most $2n$ with $l = \pm k$. The all-mode bound thus gives

$$\left|\mathbb{E}\frac{1}{n}\sum_k W(k)^2 - 1\right| + \mathbb{E}\left(\frac{1}{n}\sum_k W(k)\right)^2 \le \frac{e^{C(D+M+2)^4}}{\sqrt{n}}.$$

By (1), their difference is $\mathbb{E}\|KW\|_2^2/n$. The grafted field has at most $1 + DM$ nodes, so (13) gives the same limiting energy one, with error $A_{1+DM}/\sqrt{n}$. Substitute in (17) to obtain (18). Expand $P(X)$ in normalized multivariate Hermite monomials. Its nonconstant coefficients have squared sum at most $\mathbb{E}P(X)^2$; the number of active multi-indices is bounded by $(M+1)4^{DM}$. Cauchy–Schwarz absorbs this number into the exponential in (19). The constant term is killed by K and every other term has positive even Hermite order, hence at least two children. □

6 The Gaussian input and transfer to the original signs

Let $\mathcal{T}$ contain the rooted trees with positive even arity at every internal vertex, including the leaf L. The map J_{tr} above extends by Gaussian Hermite orthogonality to an isometry from centered, globally even Gaussian functions onto the closed linear span of the nonleaf coordinates. This defines all objects in the following input, independently of a modulus.

Lemma 6.1 (Gaussian input from [3])**.** *Fix $\tau > 0$ and write $\phi_\tau(x) = \sqrt{x^2 + 4\tau}$. There is a Gaussian linear combination $H_\tau = \sum_{T\in\mathcal{T}} c_T G_T$ with variance $Q_\tau \in [4, 2\pi]$ such that*

$$H_\tau = 2G_L + J_{\mathrm{tr}}(\phi_\tau(H_\tau) - \mathbb{E}\phi_\tau(H_\tau)). \tag{20}$$

For $X_D = \sum_{d(T)\le D} c_T G_T$ and $q_D = \mathbb{E}X_D^2$,

$$4 \le q_D \le 2\pi, \qquad \|H_\tau - X_D\|_{L^2}^2 \le C_\tau e^{-c_\tau D^{1/3}}.$$

If $P_{M,q}$ is the $L^2(N(0,q))$ projection of ϕ_τ onto polynomials of degree at most M, it is even and, for $q \in [4, 2\pi]$,

$$\mathbb{E}|\phi_\tau(\sqrt{q}Z) - P_{M,q}(\sqrt{q}Z)|^2 \le C_\tau e^{-c_\tau\sqrt{M}}, \qquad Z \sim N(0,1).$$

These are [3, Theorem 9.3, Proposition 9.4 and Theorem 9.5]. In particular their degree and Hermite tails are quantitative statements, and will be used at growing cutoffs.

We also need the scalar transfer estimate

$$\frac{1}{n}\sum_{k}\mathbb{E}|P_{M,q_D}(X_D[n](k))-\phi_\tau(X_D[n](k))|^2\le C_\tau e^{-c_\tau\sqrt{M}}+e^{C_\tau DM\log(DM+2)}n^{-1/6}, \tag{21}$$

valid when $\sqrt{n}\ge 20A_D$. Here $X_D[n]=\sum_{d(T)\le D}c_T Z_T$. We specify why [3, Proposition 10.3] gives precisely this bound for the modified fields. Its compact-test argument uses the one-mode characteristic-function estimate (12), sign hypercontractivity, and bounds for the coefficients of $P_{M,q}$. All three are unchanged on $k\notin\{0,h\}$. Fourier smoothing and cutoff truncation give the error $e^{C_\tau DM\log(DM+2)}n^{-1/6}$ there, exactly as in that proposition. At either exceptional mode, (14) and hypercontractivity bound the polynomial-growth test by $e^{C_\tau DM\log(DM+2)}$. Their total contribution to the spatial average is at most twice this expression divided by n, absorbed in (21). Thus the same scalar estimate holds at growing D,M, after increasing its constant.

Proposition 6.2. *For each fixed $\tau>0$, there are deterministic $D_n=O_\tau((\log\log n)^3)$ such that $v_n=X_{D_n}[n]/\sqrt{n}$ satisfies, along even orders,*

$$\mathbb{E}\|v_n\|_2^2\longrightarrow Q_\tau, \tag{22}$$

$$(\mathbb{E}\|v_n-K\Phi_\tau(v_n)\|_2^2)^{1/2}=O_\tau((\log n)^{-4}), \tag{23}$$

$$\mathbb{E}|(v_n)_0|^2\le n^{-1+o_\tau(1)}. \tag{24}$$

Proof. For general cutoffs D,M, let $P=P_{M,q_D}$ and let $Y[n]$ represent $Y=J_{\mathrm{tr}}(P(X_D)-\mathbb{E}P(X_D))$. The function ϕ_τ is 1-Lipschitz, and centering and J_{tr} are contractions. Lemma 6.1 therefore gives

$$\|X_D-2G_L-Y\|_{L^2}^2\le C_\tau(e^{-c_\tau D^{1/3}}+e^{-c_\tau\sqrt{M}}).$$

This Gaussian linear combination contains trees of at most $1+DM$ nodes. By (13), its canonical representative $X_D[n]-2Z_L-Y[n]$, divided by $\sqrt{n}$, has expected squared norm at most the preceding bound plus $e^{C_\tau[(D+1)(M+1)]^4}/\sqrt{n}$. Also $\mathbb{E}P(X_D)^2\le\mathbb{E}\phi_\tau(X_D)^2=q_D+4\tau$, so (19) controls $(Y[n]-KP(X_D[n]))/\sqrt{n}$ by the same finite-size error. Equation (21) and $\|K\|\le 1$ control the remaining bulk activation error.

Two elementary corrections finish the calculation. Write $v=X_D[n]/\sqrt{n}$, $z=v_0$ and

$$b_n=\sqrt{z^2+4}-\sqrt{z^2+4\tau/n}-2.$$

Then $|b_n|\le 2|z|+2\sqrt{\tau/n}$, and

$$K\Phi_\tau(v)=\frac{K\phi_\tau(X_D[n])}{\sqrt{n}}+2Ke_0+b_nKe_0.$$

Unlike the full Fourier field, our leaf omits the singleton:

$$Z_L=\frac{Fa}{\sqrt{n}}-\frac{\epsilon_h\chi_h}{\sqrt{n}},\qquad \chi_h(k)=(-1)^k,\qquad 2Ke_0-\frac{2Z_L}{\sqrt{n}}=\frac{2\epsilon_h\chi_h}{n}. \tag{25}$$

The last vector has squared Euclidean norm exactly $4/n$. We consequently have the complete residual decomposition

$$\begin{aligned}v-K\Phi_\tau(v)=&\frac{X_D[n]-2Z_L-Y[n]}{\sqrt{n}}+\frac{Y[n]-KP(X_D[n])}{\sqrt{n}}\\&+\frac{K(P(X_D[n])-\phi_\tau(X_D[n]))}{\sqrt{n}}-b_nKe_0-\frac{2\epsilon_h\chi_h}{n}.\end{aligned}$$

By (14), $\mathbb{E}|z|^2 \le q_D D! 2^{3D+2}/n$. The squared norm of the sum of the five vectors is at most five times the sum of their squared norms. For $\sqrt{n} \ge 20A_D$ this proves

$$\begin{aligned}\mathbb{E}\|v - K\Phi_\tau(v)\|_2^2 \le C_\tau(e^{-c_\tau D^{1/3}} + e^{-c_\tau\sqrt{M}}) \\ + e^{C_\tau[(D+1)(M+1)]^4} n^{-1/6}.\end{aligned} \tag{26}$$

Choose $D_n = \lceil B_\tau(\log\log n)^3 \rceil$ and an even $M_n \ge B'_\tau(\log\log n)^2$ of that order, with sufficiently large fixed constants. The first two terms are $O_\tau((\log n)^{-8})$. All exponential finite-size factors are $n^{o_\tau(1)}$ and the threshold $\sqrt{n} \ge 20A_{D_n}$ eventually holds. This proves (23). Equation (13), $q_{D_n} \to Q_\tau$, and (14) prove (22) and (24). Arbitrary fixed choices handle the finitely many smaller orders. □

7 Expectation closure

Fix $0 < \tau \le 1$ and use the vectors of Proposition 6.2. Let $\mathcal{E}_n$ be the simultaneous event in Lemma 3.1. The normalized positive vectors $p^*/\sqrt{nW}$ and $q^*/\sqrt{nW}$ have ℓ_1 norm one and satisfy the two kernel conditions. On $\mathcal{E}_n$,

$$\|f^*\|_2 \le 2C\sqrt{W}(\log n)^{3/2}.$$

With $r = v - K\Phi_\tau(v)$ define the deterministic quantity

$$Z_n = \mathbb{E}|v_0|^2 + \left(\sqrt{\mathbb{E}\|v\|_2^2} + C\sqrt{W}(\log n)^{3/2}\right)\sqrt{\mathbb{E}\|r\|_2^2}.$$

The preceding proposition implies $Z_n\sqrt{\log n} \to 0$. From (5) and Cauchy–Schwarz, $\mathbb{E}[\Delta(v)\mathbf{1}_{\mathcal{E}_n}] \le Z_n$. Take

$$\delta_n = \max\{(Z_n\sqrt{\log n})^{1/4}, (\log n)^{-1}\}.$$

The probability p_n of

$$\mathcal{E}_n^c \cup \{|v_0| > \delta_n\} \cup (\mathcal{E}_n \cap \{\Delta(v) > \delta_n^2\})$$

is at most $e^{-c(\log n)^2} + Z_n/\delta_n^2$, and therefore $p_n\sqrt{\log n} \to 0$. Outside this event, the certificate and the bound $|\operatorname{arsinh} t| \le |t|$ give

$$\frac{\theta(G_n)}{\sqrt{n}} \le \sqrt{1+\tau}\, e^{3\delta_n/2}.$$

The contribution of the event itself tends to zero by (8); for example $p\sqrt{\log(n/p)} \le p\sqrt{\log n} + p\sqrt{\log(1/p)}$. Thus the even-order upper limit is at most $\sqrt{1+\tau}$. Let $\tau \downarrow 0$ and use (4) to prove the first assertion of Theorem 1.1. Combining it with [3, Theorem 1.1] proves the all-integer assertion.

8 Exact auxiliary verification

The accompanying standard-library Python program `verify_even_projection.py` represents a Fourier–Walsh polynomial by its integer coefficients indexed by a cyclic frequency and a sign mask. Direct enumeration of admissible leaf assignments is compared with convolution of the child polynomials, application of the full Fourier multiplier and projection onto degree d with the singleton sign excluded. Lower Hermite terms are included; their normalizing powers of n are factored out exactly.

The program checks (16), (17), $K^2 = I - J$ and (25) for $n = 4, 6, 8, 10, 12, 16, 18, 20, 24$. The grafted trees are a two-leaf root, a four-leaf root, a root with one leaf and one binary child, and a root with two binary children. All 36 cases pass; 20 have nonzero canonical fields, and singleton-containing defect terms occur. These finite identities provide a reproducible check of the parity modifications. The asymptotic estimates are the analytic argument above.

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